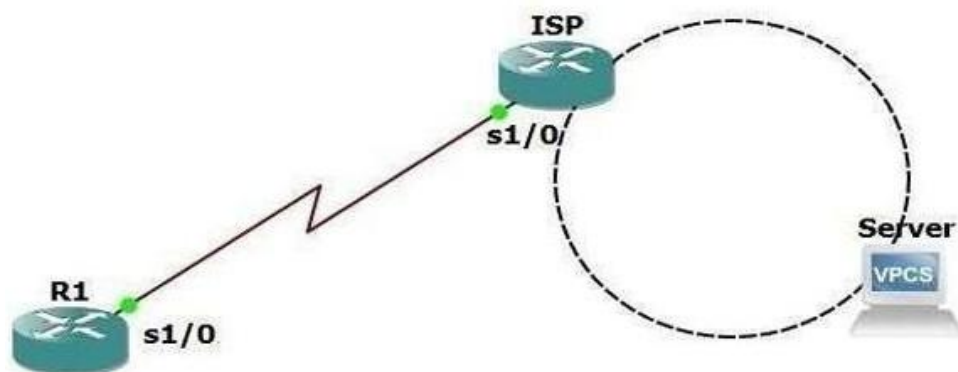


INDEX

No	Detail	Page No	Date	Sign
1	Configure IP SLA Tracking and Path Control	1		
2	CCNP ROUTE : Using the AS_PATH Attribute (Version 7)	5		
3	Configuring IBGP and EBGP Sessions, Local Preference and MED	11		
4	Secure the Management Plane	20		
5	Configure and Verify Path Control Using PBR	27		
6	Configuring remote SPAN	33		
7	Configuring Inter-VLAN Routing	35		
8 A	Configuring MPLS	43		
8 B	Implement VRF	47		

Practical No: 1

Practical 1: Configure IP SLA (On GNS3).



Device	Interface	IP address	Network Mask
R1	s1/0	209.165.200.9	255.255.255.252
ISP	s1/0	209.165.200.10	255.255.255.252
	Lo 0	198.133.209.1	255.255.255.255

```
R1#conf t
R1(config)#int s1/0
R1(config-if)#ip add 209.165.200.9 255.255.255.252
R1(config-if)#no shut
R1(config-if)#ip route
R1(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.200.10
R1(config)#exit
```

```
R2#conf t
R2(config)#hostname ISP
ISP(config)#int s1/0
ISP(config-if)#ip add 209.165.200.10 255.255.255.252
ISP(config-if)#clock rate 4032000
ISP(config-if)#exit
ISP(config)#no ip domain-lookup
ISP(config)#int loopback 0
ISP(config-if)#ip add 198.133.219.1 255.255.255.255
ISP(config-if)#no shut
```

```
ISP(config-if)#exit
ISP(config)#int s1/0
ISP(config-if)#no shut
ISP(config-if)#exit
```

```
ISP#ping 198.133.219.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 198.133.219.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/4 ms
ISP#
```

```
R1#ping 209.165.200.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/64/76 ms
R1#ping 198.133.219.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 198.133.219.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 68/68/72 ms
R1#
```

```
R1#conf t
R1(config)#ip sla 22
R1(config-ip-sla)#icmp-echo 198.133.219.1
R1(config-ip-sla-echo)#frequency 20
R1(config-ip-sla-echo)#ip sla schedule 22 start-time now life forever
R1(config)#end
```

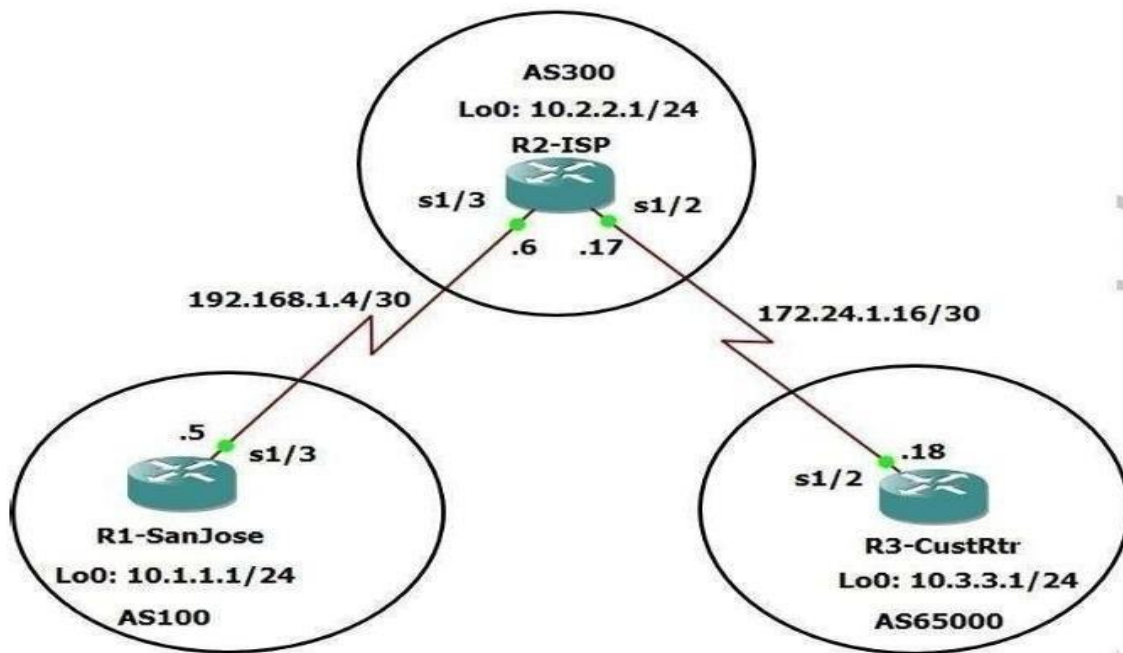
```
R1
R1#sh ip sla configuration
IP SLAs Infrastructure Engine-II
Entry number: 22
Owner:
Tag:
Type of operation to perform: icmp-echo
Target address/Source address: 198.133.219.1/0.0.0.0
Operation timeout (milliseconds): 5000
Type Of Service parameters: 0x0
Vrf Name:
Request size (ARR data portion): 28
Verify data: No
Schedule:
  Operation frequency (seconds): 20 (not considered if randomly scheduled)
  Next Scheduled Start Time: Start Time already passed
  Group Scheduled : FALSE
  Randomly Scheduled : FALSE
  Life (seconds): Forever
  Entry Ageout (seconds): never
  Recurring (Starting Everyday): FALSE
  Status of entry (SNMP RowStatus): Active
Threshold (milliseconds): 5000
Distribution Statistics:
  Number of statistic hours kept: 2
```

```
ISP#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ISP(config)#int loopback 0
ISP(config-if)#shutdown
ISP(config-if)#
*Apr 28 19:00:27.807: %LINK-5-CHANGED: Interface Loopback0, changed state to administratively down
*Apr 28 19:00:28.807: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to down
ISP(config-if)#no shut
ISP(config-if)#
*Apr 28 19:00:41.215: %LINK-3-UPDOWN: Interface Loopback0, changed state to up
*Apr 28 19:00:42.215: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
ISP(config-if)#
```

```
***  
R1#sh ip sla statistics  
  
Round Trip Time (RTT) for      Index 22  
    Latest RTT: 40 milliseconds  
Latest operation start time: *19:08:17.135 UTC Thu Apr 28 2022  
Latest operation return code: OK  
Number of successes: 79  
Number of failures: 0  
Operation time to live: Forever
```

Practical 2

Practical 2: Using the AS_PATH Attribute (On GNS3)



R1#conf t

R1(config)#hostname R1-SanJose R1-SanJose(config)#int lo 0

R1-SanJose(config-if)#ip add 10.1.1.1 255.255.255.0

R1-SanJose(config-if)#no shut

R1-SanJose(config-if)#int se1/3

R1-SanJose(config-if)#ip add 192.168.1.5 255.255.255.252

R1-SanJose(config-if)#no shut

R1-SanJose(config-if)#exit

R2#conf t

R2-ISP(config)#int lo 0

R2-ISP(config-if)#ip add 10.2.2.1 255.255.255.0

R2-ISP(config-if)#no shut

R2-ISP(config-if)#int se1/3

R2-ISP(config-if)#ip add 192.168.1.6 255.255.255.252

R2-ISP(config-if)#no shut

R2-ISP(config-if)#int se1/2

R2-ISP(config-if)#ip add 172.24.1.17 255.255.255.252

R2-ISP(config-if)#no shut

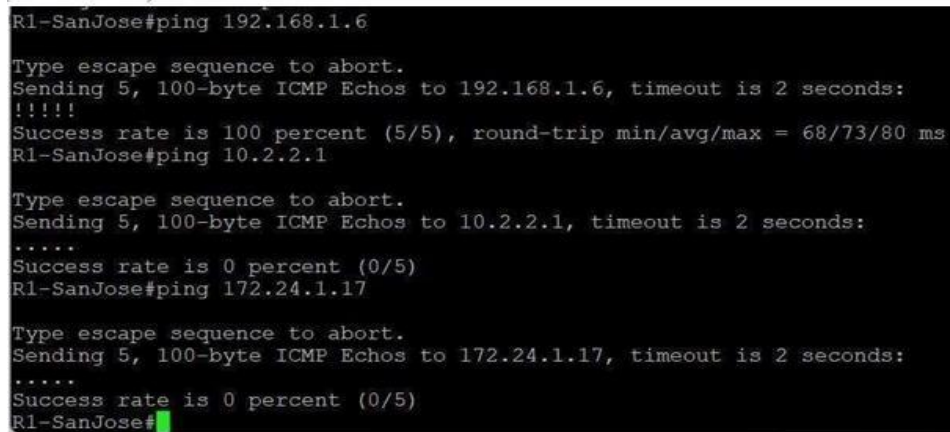
R2-ISP(config-if)#exit

R2-ISP(config)#exit

```

R3#conf t
R3(config)#hostname R3-CustRtr
R3-CustRtr(config)#int lo 0
R3-CustRtr(config-if)#ip add 10.3.3.1 255.255.255.0
R3-CustRtr(config-if)#no shut
R3-CustRtr(config-if)#int se1/2
R3-CustRtr(config-if)#ip add 172.24.1.18 255.255.255.252
R3-CustRtr(config-if)#no shut
R3-CustRtr(config-if)#exit
R3-CustRtr(config)#exit

```



```

R1-SanJose#ping 192.168.1.6
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 68/73/80 ms
R1-SanJose#ping 10.2.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1-SanJose#ping 172.24.1.17
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.24.1.17, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1-SanJose#

```

```

R1-SanJose#conf t
R1-SanJose(config)#router bgp 100
R1-SanJose(config-router)#neighbor 192.168.1.6 remote-as 300
R1-SanJose(config-router)#network 10.1.1.0 mask 255.255.25.0

```

```

R2-ISP#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2-ISP(config)#router bgp 300
R2-ISP(config-router)#neighbor 192.168.1.5 remote-as 100
R2-ISP(config-router)#%BGP-5-ADJCHANGE: neighbor 192.168.1.5 Up
R2-ISP(config-router)#neighbor 172.24.1.18 remote-as 65000
R2-ISP(config-router)#network 10.2.2.0 mask 255.255.255.0

```

```

R3-CustRtr#conf t
R3-CustRtr(config)#router bgp 65000
R3-CustRtr(config-router)#neighbor 172.24.1.17 remote-as 300
R3-CustRtr(config-router)#%BGP-5-ADJCHANGE: neighbor 172.24.1.17 Up
R3-CustRtr(config-router)#network 10.3.3.0 mask 255.255.255.0

```



```
R2-ISP#sh ip bgp neighbors
BGP neighbor is 172.24.1.18, remote AS 65000, external link
BGP version 4, remote router ID 10.3.3.1
BGP state = Established, up for 00:11:04
```

```
BGP neighbor is 192.168.1.5, remote AS 100, external link
BGP version 4, remote router ID 10.1.1.1
BGP state = Established, up for 00:15:06
```

```
R1-SanJose#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/24 is subnetted, 3 subnets
B    10.3.3.0 [20/0] via 192.168.1.6, 00:39:54
B    10.2.2.0 [20/0] via 192.168.1.6, 00:42:36
C    10.1.1.0 is directly connected, Loopback0
    192.168.1.0/30 is subnetted, 1 subnets
C    192.168.1.4 is directly connected, Serial1/3
R1-SanJose#
```

```
R1-SanJose#ping 10.3.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.3.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1-SanJose#
```

```
R1-SanJose#ping
Protocol [ip]:
Target IP address: 10.3.3.1
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [3]:
Extended commands [n]: y
Source address or interface: 10.1.1.1
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.1.1.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/63/88 ms
R1-SanJose#ping 10.3.3.1 source 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.1.1.1
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/66/96 ms
R1-SanJose#
```

```
R1-SanJose#sh ip bgp
BGP table version is 4, local router ID is 10.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop        Metric LocPrf Weight Path
*> 10.1.1.0/24     0.0.0.0          0           32768 i
*> 10.2.2.0/24     192.168.1.6      0           0 300 i
*> 10.3.3.0/24     192.168.1.6      0           0 300 65000 i
R1-SanJose#
```

R2-ISP#conf t

R2-ISP(config)#router bgp 300

R2-ISP(config-router)#neighbor 192.168.1.5 remove-private-as

R2-ISP(config-router)#^Z

```
R2-ISP#clear ip bgp *
R2-ISP#
*Apr 28 20:55:26.451: %BGP-5-ADJCHANGE: neighbor 172.24.1.18 Down User reset
*Apr 28 20:55:26.455: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Down User reset
R2-ISP#
*Apr 28 20:55:28.511: %BGP-5-ADJCHANGE: neighbor 172.24.1.18 Up
*Apr 28 20:55:28.523: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Up
R2-ISP#
```

```
R1-SanJose#ping 10.3.3.1 source 10 0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.1.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/59/88 ms
R1-SanJose#
```

```
R1-SanJose#ping 10.3.3.1 source 10 0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.1.1.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/59/88 ms
R1-SanJose#
```

R2-ISP#conf t

R2-ISP(config)#ip as-path access-list 1 deny ^100\$

R2-ISP(config)#ip as-path access-list 1 permit .*

R2-ISP#conf t

R2-ISP(config)#router bgp 300

R2-ISP(config-router)#neighbor 172.24.1.18 filter-list 1 out

R2-ISP(config-router)#^Z R2-ISP#

```
R2-ISP#clear ip bgp *
R2-ISP#
*Apr 28 21:12:51.315: %BGP-5-ADJCHANGE: neighbor 172.24.1.18 Down User reset
*Apr 28 21:12:51.315: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Down User reset
R2-ISP#
*Apr 28 21:12:52.599: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Up
*Apr 28 21:12:53.375: %BGP-5-ADJCHANGE: neighbor 172.24.1.18 Up
R2-ISP#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  172.24.0.0/30 is subnetted, 1 subnets
C       172.24.1.16 is directly connected, Serial1/2
  10.0.0.0/24 is subnetted, 2 subnets
B       10.3.3.0 [20/0] via 172.24.1.18, 00:00:17
C       10.2.2.0 is directly connected, Loopback0
  192.168.1.0/30 is subnetted, 1 subnets
C       192.168.1.4 is directly connected, Serial1/3
R2-ISP#
```

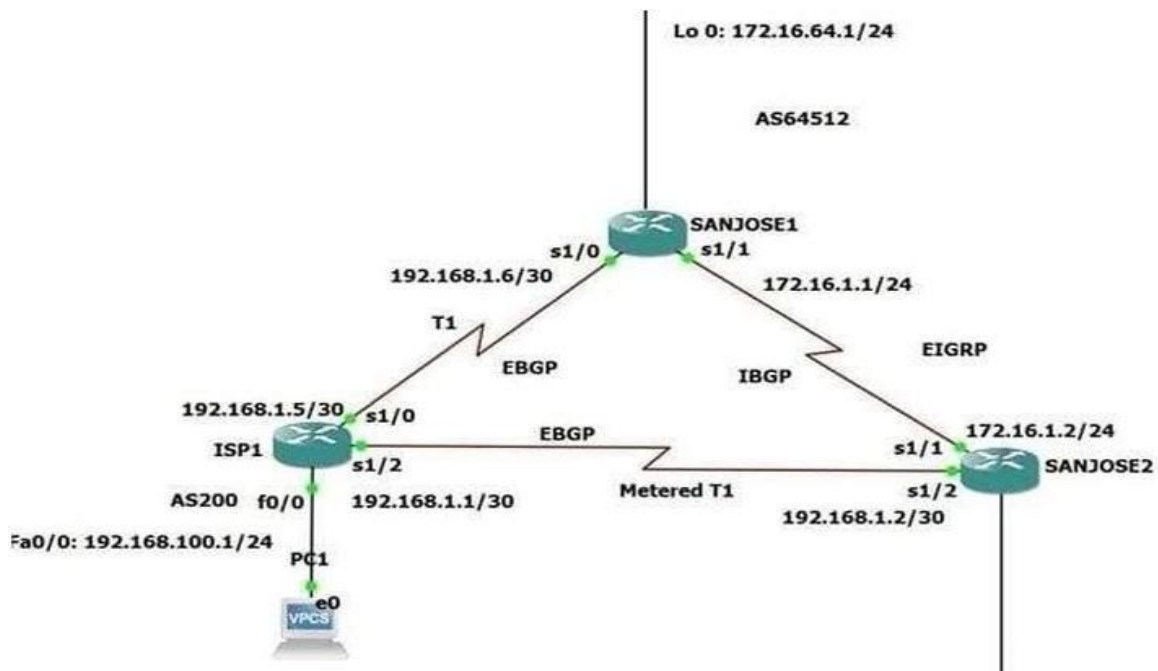
```
R3-CustRtr#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.24.0.0/30 is subnetted, 1 subnets
C       172.24.1.16 is directly connected, Serial1/2
    10.0.0.0/24 is subnetted, 2 subnets
C       10.3.3.0 is directly connected, Loopback0
B       10.2.2.0 [20/0] via 172.24.1.17, 00:03:19
R3-CustRtr#
```

Practical 3

Practical 3: Configuring IBGP and EBGp Sessions, Local Preferences and MED



```
ISP1#conf t
ISP1(config)#int s1/0
ISP1(config-if)#ip add 192.168.1.5 255.255.255.252
ISP1(config-if)#no shut
ISP1(config-if)#int s1/2
ISP1(config-if)#ip add 192.168.1.1 255.255.255.252
ISP1(config-if)#ip add 192.168.1.1 255.255.255.252
ISP1(config-if)#no shut
ISP1(config-if)#int fa0/0
ISP1(config-if)#ip add 192.168.100.1 255.255.255.252
ISP1(config-if)#ip add 192.168.100.1 255.255.255.0
ISP1(config-if)#no shut
```

```
SANJOSE1#conf t
SANJOSE1(config)#int s1/0
SANJOSE1(config-if)#ip add 192.168.1.6 255.255.255.252
SANJOSE1(config-if)#no shut
SANJOSE1(config-if)#int s1/1
SANJOSE1(config-if)#ip add 172.16.1.1 255.255.255.0
SANJOSE1(config-if)#no shut
SANJOSE1(config-if)#int lo 0
SANJOSE1(config-if)#ip add 172.16.64.1 255.255.255.0
```

SANJOSE1(config-if)#no shut

SANJOSE2#conf t

SANJOSE2(config)#int s1/1

SANJOSE2(config-if)#ip add 172.16.1.2 255.255.255.0

SANJOSE2(config-if)#no shut

SANJOSE2(config-if)#int s1/2

SANJOSE2(config-if)#ip add 172.168.1.2 255.255.255.252

SANJOSE2(config-if)#no shut

SANJOSE2(config-if)#int lo 0

SANJOSE2(config-if)#ip add 172.16.32.1 255.255.255.0

SANJOSE2(config-if)#no shut

SANJOSE1#conf t

SANJOSE1(config)#router eigrp 64512

SANJOSE1(config-router)#network 172.16.0.0

SANJOSE1(config-router)#exit

SANJOSE2#conf t

SANJOSE2(config)#router eigrp 64512

SANJOSE2(config-router)#network 172.16.0.0

SANJOSE2(config-router)#exit

SANJOSE1#conf t

SANJOSE1(config)#router bgp 64512

SANJOSE1(config-router)#no auto-summary

SANJOSE1(config-router)#neighbor 172.16.32.1 update-source lo 0

SANJOSE1#conf t

SANJOSE1(config)#router bgp 64512

SANJOSE1(config-router)#no synchronization

SANJOSE2#conf t

SANJOSE2(config)#router bgp 64512

SANJOSE2(config-router)#no synchronization

SANJOSE2#conf t

SANJOSE2(config)#router bgp 64512

SANJOSE2(config-router)#no synchronization

SANJOSE2(config-router)#neighbor 172.16.64.1 remote-as 64512

SANJOSE2(config-router)#neighbor 172.16.64.1 update-source lo0


```

SANJOSE1
Enter configuration commands, one per line. End with CNTL/Z.
SANJOSE1(config)#router bgp 64512
SANJOSE1(config-router)#no synchronization
SANJOSE1(config-router)#
SANJOSE1#
*Jun 24 10:07:54.839: %SYS-5-CONFIG_I: Configured from console by console
SANJOSE1#
SANJOSE1#
SANJOSE1#sh ip bgp neighbors
BGP neighbor is 172.16.32.1, remote AS 64512, internal link
  BGP version 4, remote router ID 0.0.0.0
  BGP state = Active
  Last read 00:13:16, last write 00:13:16, hold time is 180, keepalive interval
  is 60 seconds
  Message statistics:
    InQ depth is 0
    OutQ depth is 0
    Opens:          Sent      Rcvd
    Notifications:  0        0
    Updates:        0        0
    Keepalives:     0        0
    Route Refresh:  0        0
    Total:          0        0
  Default minimum time between advertisement runs is 0 seconds

For address family: IPv4 Unicast
  BGP table version 1, neighbor version 0/0
  Output queue size : 0
  Index 1, Offset 0, Mask 0x2
  1 update-group member
    Prefix activity:
      Sent      Rcvd
      Prefixes Current: 0      0
      Prefixes Total:   0      0
      Implicit Withdraw: 0      0
      Explicit Withdraw: 0      0
      Used as bestpath:  n/a    0
      Used as multipath: n/a    0
    Local Policy Denied Prefixes:
      Outbound  Inbound
      Total:    0      0
  Number of NLRs in the update sent: max 0, min 0
  Connections established 0; dropped 0
  Last reset never
  No active TCP connection
SANJOSE1#
SANJOSE1#
SANJOSE1#
SANJOSE1#

```

```

SANJOSE2
Invalid input detected at '^' marker.
SANJOSE2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SANJOSE2(config)#router bgp 64512
SANJOSE2(config-router)#no synchronization
SANJOSE2(config-router)#
SANJOSE2(config-router)#
SANJOSE2#
*Jun 24 10:03:09.379: %SYS-5-CONFIG_I: Configured from console by console
SANJOSE2#
SANJOSE2#
SANJOSE2#
SANJOSE2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SANJOSE2(config)#router bgp 64512
SANJOSE2(config-router)#no synchronization
SANJOSE2(config-router)#neighbor 172.16.64.1 remote-as 64512
SANJOSE2#
Invalid input detected at '^' marker.
SANJOSE2(config-router)#neighbor 172.16.64.1 remote-as 64512
SANJOSE2(config-router)#neighbor 172.16.64.1 update-source lo0
SANJOSE2(config-router)#^Z
SANJOSE2#
*Jun 24 10:06:44.239: %SYS-5-CONFIG_I: Configured from console by console
SANJOSE2#
SANJOSE2#
SANJOSE2#sh ip bgp neighbors
BGP neighbor is 172.16.64.1, remote AS 64512, internal link
  BGP version 4, remote router ID 0.0.0.0
  BGP state = Active
  Last read 00:04:01, last write 00:04:01, hold time is 180, keepalive interval
  is 60 seconds
  Message statistics:
    InQ depth is 0
    OutQ depth is 0
    Opens:          Sent      Rcvd
    Notifications:  0        0
    Updates:        0        0
    Keepalives:     0        0
    Route Refresh:  0        0
    Total:          0        0
  Default minimum time between advertisement runs is 0 seconds

For address family: IPv4 Unicast
  BGP table version 1, neighbor version 0/0
  Output queue size : 0
  Index 1, Offset 0, Mask 0x2
  1 update-group member
    Sent      Rcvd

```

```

ISP1#ping 172.16.64.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.64.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/31/44 ms
ISP1#ping 172.16.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/29/32 ms
ISP1#

```

```
ISP1(config-router)#network 192.168.100.0
SANJOSE1#conf t
SANJOSE1(config)#ip route 172.16.0.0 255.255.0.0 null0
SANJOSE1(config)#router bgp 64512
```

```
SANJOSE1(config-router)#neighbor 192.168.1.5 remote-as 200
SANJOSE1(config-router)#network 172.16.0.0
```

```
BGP neighbor is 192.168.1.5, remote AS 200, external link
BGP version 4, remote router ID 192.168.100.1
BGP state = Established, up for 00:01:00
```

```
SANJOSE2#conf t
SANJOSE2(config)#ip route 172.16.0.0 255.255.0.0 null0
SANJOSE2(config)#router bgp 64512
SANJOSE2(config-router)#neighbor 192.168.1.1 remote-as 200
SANJOSE2(config-router)#network 172.16.0.0
```

```
ISP1#ping 172.16.32.1
```

```
TySANJOSE2#show ip bgp summary
SeBGP router identifier 172.16.32.1, local AS number 64512
  BGP table version is 3, main routing table version 3
  *1 network entries using 144 bytes of memory
Su2 path entries using 160 bytes of memory
IS2/1 BGP path/bestpath attribute entries using 272 bytes of memory
  0 BGP route-map cache entries using 0 bytes of memory
Ty0 BGP filter-list cache entries using 0 bytes of memory
SeBGP using 576 total bytes of memory
  BGP activity 1/0 prefixes, 2/0 paths, scan interval 60 secs
..
SuNeighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
IS172.16.64.1    4      64512    15     15      3    0   0 00:09:54      1
IS192.168.1.1    4      200      6      6      3    0   0 00:00:42      0
BGSANJOSE2#
```

```
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found
```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*	172.16.0.0	192.168.1.2	0		0	64512 i
*>		192.168.1.6	0		0	64512 i
*>	192.168.100.0	0.0.0.0	0		32768	i

```
ISP1#ping
Protocol [ip]:
Target IP address: 172.16.64.1
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
```


Source address or interface: 192.168.100.1

Type of service [0]:

Set DF bit in IP header? [no]:

Validate reply data? [no]:

Data pattern [0xABCD]:

Loose, Strict, Record, Timestamp, Verbose [none]:

Sweep range of sizes [n]:

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 172.16.64.1, timeout is 2 seconds:

Packet sent with a source address of 192.168.100.1

Success rate is 100 percent (5/5), round-trip min/avg/max = 20/27/36 ms

ISP1#conf t

ISP1 (config)#router bgp 200

```
ISP1#ISP1 (config-router)#network 192.168.1.0 mask 255.255.255.252
BGP table version is 5, local router ID is 192.168.100.1
ISP1 (config-router)#network 192.168.1.4 mask 255.255.255.252
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*>  172.16.0.0       192.168.1.2             0           0 64512 i
*    172.16.1.0/24   192.168.1.6             0           0 64512 i
*>  192.168.1.0/30    0.0.0.0                 0           32768 i
*>  192.168.1.4/30    0.0.0.0                 0           32768 i
*>  192.168.100.0     0.0.0.0                 0           32768 i
ISP1#
```

SANJOSE2#sh ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override
```

Gateway of last resort is not set

```
172.16.0.0/16 is variably subnetted, 6 subnets, 3 masks
S    172.16.0.0/16 is directly connected, Null0
C    172.16.1.0/24 is directly connected, Serial1/1
L    172.16.1.2/32 is directly connected, Serial1/1
C    172.16.32.0/24 is directly connected, Loopback0
L    172.16.32.1/32 is directly connected, Loopback0
D    172.16.64.0/24 [90/2297856] via 172.16.1.1, 00:36:15, Serial1/1
    192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C    192.168.1.0/30 is directly connected, Serial1/2
L    192.168.1.2/32 is directly connected, Serial1/2
B    192.168.1.4/30 [20/0] via 192.168.1.1, 00:01:02
B    192.168.100.0/24 [20/0] via 192.168.1.1, 00:01:02
SANJOSE2#
```

```

SANJOSE1#conf t
SANJOSE1 (config)#router bgp 64512
SANJOSE1 (config-router)#neighbor 172.16.32.1 next-hop-self Tha
SANJOSE2#conf t
SANJOSE2 (config)#router bgp 64512
SANJOSE2 (config-router) #neighbor 172.16.64.1 next-hop-self

```

```

SANJOSE2#clear ip bgp *
SANJOSE2#
*May 15 09:49:46.755: %BGP-5-ADJCHANGE: neighbor 172.16.64.1 Down User reset
*May 15 09:49:46.759: %BGP_SESSION-5-ADJCHANGE: neighbor 172.16.64.1 IPv4 Unicast topology base removed from session User reset
*May 15 09:49:46.763: %BGP-5-ADJCHANGE: neighbor 192.168.1.1 Down User reset
*May 15 09:49:46.767: %BGP_SESSION-5-ADJCHANGE: neighbor 192.168.1.1 IPv4 Unicast topology base removed from session User reset
*May 15 09:49:47.375: %BGP-5-ADJCHANGE: neighbor 172.16.64.1 Up
*May 15 09:49:47.375: %BGP-5-ADJCHANGE: neighbor 192.168.1.1 Up
SANJOSE2#

```

```

SANJOSE2#show ip bgp
BGP table version is 1, local router ID is 172.16.32.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop              Metric LocPrf Weight Path
*    172.16.0.0      0.0.0.0                  0         32768 i
* i    172.16.64.1    172.16.64.1              0         100      0 i
*    192.168.1.0/30  192.168.1.1              0         100      0 200 i
* i    172.16.64.1    172.16.64.1              0         100      0 200 i
*    192.168.1.4/30  192.168.1.1              0         100      0 200 i
* i    172.16.64.1    172.16.64.1              0         100      0 200 i
*    192.168.100.0   192.168.1.1              0         100      0 200 i
* i    172.16.64.1    172.16.64.1              0         100      0 200 i
SANJOSE2#

```

```

SANJOSE1#conf t
SANJOSE1 (config)#route-map PRIMARY_T1_IN permit 10
SANJOSE1 (config-route-map) #set local-preference 150
SANJOSE1 (config)#router bgp 64512
SANJOSE1 (config-router)#neighbor 192.168.1.5 route-map PRIMARY_T1_IN i
SANJOSE2#conf t
SANJOSE2 (config)#route-map SECONDARY_T1_IN permit 10
SANJOSE2 (config-route-map) #set local-preference 125
SANJOSE2 (config-route-map) #router bgp 64512
SANJOSE2 (config-router)#neighbor 192.168.1.1 route-map SECONDARY_T1_IN
in

```



```

SANJOSE1#clear ip bgp *
SANJOSE1#
*May 15 09:55:02.459: %BGP-5-ADJCHANGE: neighbor 172.16.32.1 Down User reset
*May 15 09:55:02.463: %BGP_SESSION-5-ADJCHANGE: neighbor 172.16.32.1 IPv4 Unicast topology base removed from session User reset
*May 15 09:55:02.467: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Down User reset
*May 15 09:55:02.467: %BGP_SESSION-5-ADJCHANGE: neighbor 192.168.1.5 IPv4 Unicast topology base removed from session User reset
*May 15 09:55:02.587: %BGP-5-ADJCHANGE: neighbor 172.16.32.1 Up
*May 15 09:55:02.615: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Up
SANJOSE1#

```

```

SANJOSE1#sh ip bgp
BGP table version is 1, local router ID is 172.16.64.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop           Metric LocPrf Weight Path
*   172.16.0.0      0.0.0.0              0         32768 i
*   i              172.16.32.1          0         100      0 i
*   192.168.1.0/30  192.168.1.5          0         150      0 200 i
*   i              172.16.32.1          0         100      0 200 i
*   192.168.1.4/30  192.168.1.5          0         150      0 200 i
*   i              172.16.32.1          0         100      0 200 i
*   192.168.100.0   192.168.1.5          0         150      0 200 i
*   i              172.16.32.1          0         100      0 200 i
SANJOSE1#

```

```

SANJOSE2#clear ip bgp *
SANJOSE2#
*May 15 09:57:34.999: %BGP-5-ADJCHANGE: neighbor 172.16.64.1 Down User reset
*May 15 09:57:34.999: %BGP_SESSION-5-ADJCHANGE: neighbor 172.16.64.1 IPv4 Unicast topology base removed from session User reset
*May 15 09:57:35.007: %BGP-5-ADJCHANGE: neighbor 192.168.1.1 Down User reset
*May 15 09:57:35.007: %BGP_SESSION-5-ADJCHANGE: neighbor 192.168.1.1 IPv4 Unicast topology base removed from session User reset
*May 15 09:57:35.399: %BGP-5-ADJCHANGE: neighbor 172.16.64.1 Up
*May 15 09:57:35.399: %BGP-5-ADJCHANGE: neighbor 192.168.1.1 Up
SANJOSE2#sh ip bgp
BGP table version is 1, local router ID is 172.16.32.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop           Metric LocPrf Weight Path
*   i 172.16.0.0     172.16.64.1          0         100      0 i
*   192.168.1.0/30  192.168.1.1          0         125      0 200 i
*   i              172.16.64.1          0         150      0 200 i
*   192.168.1.4/30  192.168.1.1          0         125      0 200 i
*   i              172.16.64.1          0         150      0 200 i
*   192.168.100.0   192.168.1.1          0         125      0 200 i
*   i              172.16.64.1          0         150      0 200 i
SANJOSE2#

```

SANJOSE1#conf t

```

SANJOSE1 (config)#route-map PRIMARY_T1_MED_OUT permit 10
SANJOSE1 (config-route-map) #set Metric 50
SANJOSE1 (config-route-map)#exit
SANJOSE1 (config)#router bgp 64512
SANJOSE1 (config-router)#neighbor 192.168.1.5 route-map PRIMARY_T1_MED_OUT
out
SANJOSE2#conf t
SANJOSE2 (config)#route-map SECONDARY_T1_MED_OUT permit 10
SANJOSE2 (config-route-map) #set Metric 75
SANJOSE2 (config-route-map)#exit
SANJOSE2 (config)#router bgp 64512
SANJOSE2(config-router)#$2.168.1.1 route-map SECONDARY_T1_MED_OUT out

```

```

SANJOSE1#clear ip bgp *
SANJOSE1#
*May 15 10:13:56.855: %BGP-5-ADJCHANGE: neighbor 172.16.32.1 Down User reset
*May 15 10:13:56.859: %BGP_SESSION-5-ADJCHANGE: neighbor 172.16.32.1 IPv4 Unicast topology base removed from session User reset
*May 15 10:13:56.863: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Down User reset
*May 15 10:13:56.867: %BGP_SESSION-5-ADJCHANGE: neighbor 192.168.1.5 IPv4 Unicast topology base removed from session User reset
*May 15 10:13:57.183: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Up
*May 15 10:13:57.183: %BGP-5-ADJCHANGE: neighbor 172.16.32.1 Up
SANJOSE1#
*May 15 10:14:05.611: %BGP-5-NBR RESET: Neighbor 172.16.32.1 reset (Peer closed the session)
*May 15 10:14:05.619: %BGP-5-ADJCHANGE: neighbor 172.16.32.1 Down Peer closed the session
*May 15 10:14:05.619: %BGP_SESSION-5-ADJCHANGE: neighbor 172.16.32.1 IPv4 Unicast topology base removed from session Peer closed the session
SANJOSE1#
*May 15 10:14:06.623: %BGP-5-ADJCHANGE: neighbor 172.16.32.1 Up
SANJOSE1#

```

```

SANJOSE2#clear ip bgp *
SANJOSE2#
*May 15 10:14:05.559: %BGP-5-ADJCHANGE: neighbor 172.16.64.1 Down User reset
*May 15 10:14:05.559: %BGP_SESSION-5-ADJCHANGE: neighbor 172.16.64.1 IPv4 Unicast topology base removed from session User reset
*May 15 10:14:05.567: %BGP-5-ADJCHANGE: neighbor 192.168.1.1 Down User reset
*May 15 10:14:05.567: %BGP_SESSION-5-ADJCHANGE: neighbor 192.168.1.1 IPv4 Unicast topology base removed from session User reset
SANJOSE2#
*May 15 10:14:06.591: %BGP-5-ADJCHANGE: neighbor 192.168.1.1 Up
*May 15 10:14:06.603: %BGP-5-ADJCHANGE: neighbor 172.16.64.1 Up
SANJOSE2#

```

```

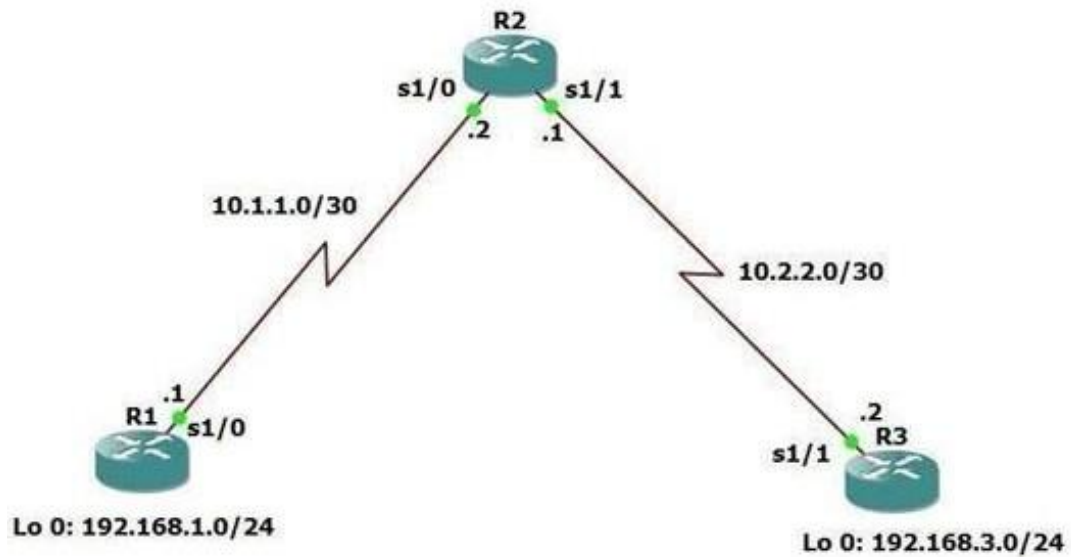
SANJOSE1#sh ip bgp
BGP table version is 5, local router ID is 172.16.64.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop        Metric LocPrf Weight Path
* i 172.16.0.0       172.16.32.1          0     100      0 i
*>          0.0.0.0          0           32768 i
*> 192.168.1.0/30    192.168.1.5          0     150      0 200 i
r> 192.168.1.4/30    192.168.1.5          0     150      0 200 i
*> 192.168.100.0     192.168.1.5          0     150      0 200 i
SANJOSE1#

```

Practical 4

Practical 4: Configuring Secure Management Plane (On GNS3)



```
R1#conf t
R1(config)#int lo 0
R1(config-if)#ip add 192.168.1.1 255.255.255.0
R1(config-if)#exit
R1(config)#int se1/0
R1(config-if)#ip add 10.1.1.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#exit
R2#conf t
R2(config)#int se1/0
R2(config-if)#ip add 10.1.1.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int se1/1
R2(config-if)#ip add 10.2.2.1 255.255.255.252
R2(config-if)#no shut
R2(config-if)#exit
R3#conf t
R3(config)#int lo 0
R3(config-if)#ip add 192.168.3.1 255.255.255.0
R3(config-if)#exit
R3(config)#int se1/1
R3(config-if)#ip add 10.2.2.2 255.255.255.252
R3(config-if)#no shut
R3(config-if)#exit
```



```
R1#conf t
R1(config)#ip route 0.0.0.0 0.0.0.0 10.1.1.2
R3#conf t
R3(config)#ip route 0.0.0.0 0.0.0.0 10.2.2.1
R2#conf t
R2(config)#ip route 192.168.1.0 255.255.255.0 10.1.1.1
R2(config)#ip route 192.168.3.0 255.255.255.0 10.2.2.2
```

```
R1#ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/8 ms
R1#ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/64/76 ms
R1#ping 10.1.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/31/48 ms
R1#ping 10.2.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/27/32 ms
R1#ping 10.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/58/60 ms
R1#ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/56/64 ms
R1#
```

```
R1(config)#line console 0
R1(config-line)#password ciscoconpass
R1(config-line)#exec-timeout 5 0
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exit
R1(config)#line vty 0 4
R1(config-line)#password ciscovtypass
R1(config-line)#exec-timeout 5 0
R1(config-line)#login
R1(config-line)#exit
R1(config)#line aux 0
```

```
R1(config-line)#no exec
```

```
R1(config-line)#end
```

```
R1#conf t
```

```
R1(config)#service password-encryption
```

```
R3#conf t
```

```
R3(config)#security passwords min-length 10
```

```
R3(config)#enable secret class12345
```

```
R3(config)#line console 0
```

```
R3(config-line)#password ciscoconpass
```

```
R3(config-line)#exec-timeout 5 0
```

```
R3(config-line)#login
```

```
R3(config-line)#logging synchronous
```

```
R3(config-line)#exit
```

```
R3(config)#line vty 0 4
```

```
R3(config-line)#password ciscovtypass
```

```
R3(config-line)#exec-timeout 5 0
```

```
R3(config-line)#login
```

```
R3(config-line)#exit
```

```
R3(config)#line aux 0
```

```
R3(config-line)#no exec
```

```
R3(config-line)#end
```

```
R3#conf t
```

```
R3(config)#service password-encryption
```

```
R3(config)#banner motd $Unauthorized access strictly prohibited!$
```

```
R3(config)#exit
```

```
R1#conf t
```

```
R1(config)#username JR-ADMIN secret class12345
```

```
R1(config)#username ADMIN secret class54321
```

```
R1(config)#line console 0
```

```
R1(config-line)#login local
```

```
R1(config-line)#exit
```

```
R1(config)#line vty 0 4
```

```
R1(config-line)#login local
```

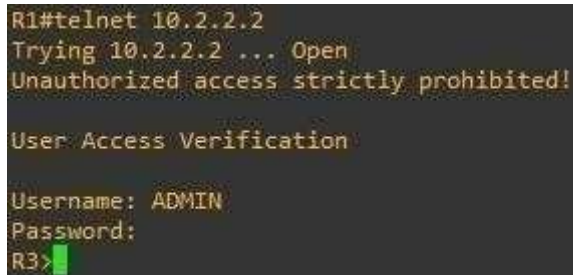
```
R1(config-line)#end R1
```

```
R3#conf t
```

```
R3(config)#username JR-ADMIN secret class12345
```

```
R3(config)#username ADMIN secret class54321
```

```
R3(config)#line console 0
R3(config-line)#login local
R3(config-line)#exit
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#end
```



```
R1#telnet 10.2.2.2
Trying 10.2.2.2 ... Open
Unauthorized access strictly prohibited!

User Access Verification

Username: ADMIN
Password:
R3>
```

```
R1(config)#radius server RADIUS-1
R1(config-radius-server)#address ipv4 192.168.1.101
R1(config-radius-server)#key RADIUS-1-pa55w0rd
R1(config-radius-server)#exit
```

```
R1(config)#radius server RADIUS-2
R1(config-radius-server)#address ipv4 192.168.1.102
R1(config-radius-server)#key RADIUS-2-pa55w0rd
R1(config-radius-server)#exit
```

```
R1(config-sg-radius)#server name RADIUS-1
R1(config-sg-radius)#server name RADIUS-2
R1(config-sg-radius)#exit
```

```
R1(config)#line vty 0 4
R1(config-line)#login authentication TELNET-LOGIN
R1(config-line)#exit
```

```
R3#conf t
R3(config)#aaa new-model
R3(config)#radius server RADIUS-1
R3(config-radius-server)#address ipv4 192.168.1.101
R3(config-radius-server)#key RADIUS-1-pa55w0rd
R3(config-radius-server)#exit
R3(config)#radius server RADIUS-2
R3(config-radius-server)#address ipv4 192.168.1.102
```

```

R3(config-radius-server)#key RADIUS-2-pa55w0rd
R3(config-radius-server)#exit
R3(config)#aaa group server radius RADIUS-GROUP
R3(config-sg-radius)#server name RADIUS-1
R3(config-sg-radius)#server name RADIUS-2
R3(config-sg-radius)#exit
R3(config)#aaa authentication login default group RADIUS-GROUP local
R3(config)#aaa authentication login TELNET-LOGIN group RADIUS-GROUP local-
case
R3(config)#line vty 0 4
R3(config-line)#login authentication TELNET-LOGIN
R3(config-line)#exit

```



```

R1#telnet 10.2.2.2
Trying 10.2.2.2 ... Open
Unauthorized access strictly prohibited!
User Access Verification

Username: admin
Password:

% Authentication failed

R1#telnet 10.2.2.2
Trying 10.2.2.2 ... Open
Unauthorized access strictly prohibited!
User Access Verification

Username: ADMIN
Password:

R3>

```

```

R1#conf t
R1(config)#ip domain-name ccnasecurity.com
R1(config-line)#transport input ssh
R1(config-line)#end

```



```

R1#sh ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQCusWpZxSxci4AXX7csxYc5winMsKCEdmk1t+PuK2aU
30msvz62cjMENZXcg582wcW6MsqNqCmQxWxeQuwt672MwSxZ9x+8EncVJsmbPKPzO4tioSi0IRbpicD7A
fUFtMiqzreuJ5U6Uhp08b9EBFJqnczLJAkUMyzDRq80cRgFOTw==
R1#

```

```

R3#conf t
R3(config)#ip domain-name ccnasecurity.com
R3(config)#crypto key zeroize rsa
R3(config)#crypto key generate rsa general-keys modulus 1024
R3(config)#ip ssh version 2

```

R3(config)#line vty 0 4

R3(config-line)#transport input ssh

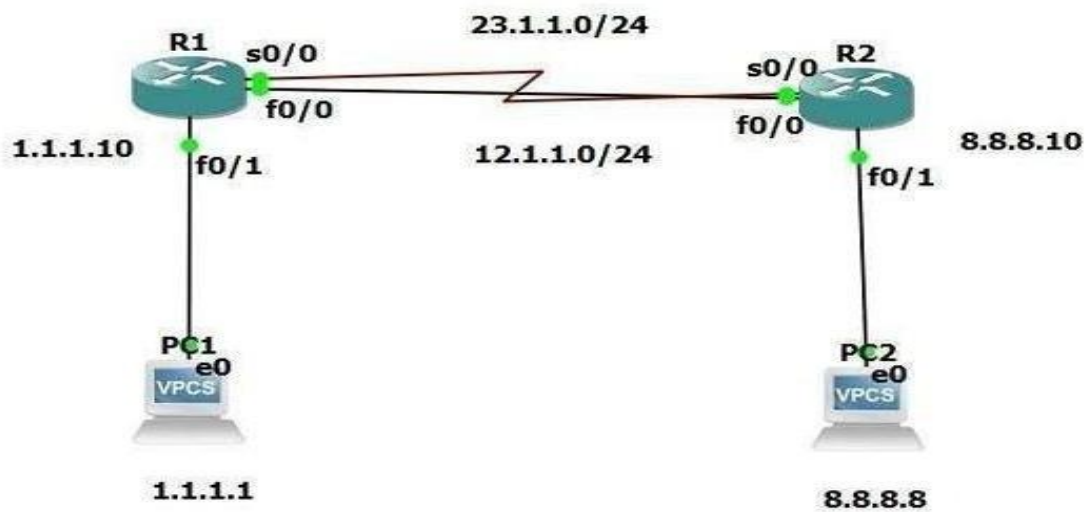
R3(config-line)#end

```
R3#sh ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGCizxUKc0w5wB/m8wbM9o0m17xXFJagVcT0wkQY3bfQ
sKai44Y6J/6ycE7ZnwUjRUOvkNXrKFUcd0BBtugSesjAxUV3LRilMpQWttab/V3klGNsZ+KaEKd8z09d
uAuXH5s+fdoPGkoDzb/xlFxRpGnDf7XNs0MsHjrWj32dp1p0Yw==
R3#
```

```
R1#ssh -l ADMIN 10.2.2.2
Password:
Unauthorized access strictly prohibited!R3>en
Password:
R3#
```


Practical 5

Practical 5: Configuring PBR (on GNS3).



```
R1#conf t
R1(config)#int fa0/0
R1(config-if)#ip add 12.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int s2/0
R1(config-if)#ip add 23.1.1.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int f1/0
R1(config-if)#ip add 1.1.1.10 255.255.255.0
R1(config-if)#no shut
R1(config)#exit
```

```
R2#conf t
R2(config)#int fa0/0
R2(config-if)#ip add 12.1.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int se2/0
R2(config-if)#ip add 23.1.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

```
R2(config)#int fa1/0
R2(config-if)#ip add 8.8.8.10 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

```
R1#conf t
R1(config)#router ospf 100
R1(config-router)#network 12.1.1.0 0.0.0.255 area 1
R1(config-router)#network 23.1.1.0 0.0.0.255 area 1
R1(config-router)#network 1.1.1.0 0.0.0.255 area 1
```

```
R2#conf t
R2(config)#router ospf 100
R2(config-router)#network 12.1.1.0 0.0.0.255 area 1
R2(config-router)#network 12.1.1.0 0.0.0.255 area 1
R2(config-router)#network 23.1.1.0 0.0.0.255 area 1
R2(config-router)#network 23.1.1.0 0.0.0.255 area 1
R2(config-router)#network 8.8.8.0 0.0.0.255 area 1
R2(config-router)#exit
R2(config)#exit
```

```
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  1.0.0.0/24 is subnetted, 1 subnets
C       1.1.1.0 is directly connected, FastEthernet1/0
  23.0.0.0/24 is subnetted, 1 subnets
C       23.1.1.0 is directly connected, Serial2/0
  8.0.0.0/24 is subnetted, 1 subnets
O       8.8.8.0 [110/2] via 12.1.1.2, 00:00:32, FastEthernet0/0
 12.0.0.0/24 is subnetted, 1 subnets
C       12.1.1.0 is directly connected, FastEthernet0/0
R1#
```

```
R2
R2 (config)#exit
R2#
*Apr 29 12:21:22.347: %SYS-5-CONFIG_I: Configured from console by console
R2#
R2#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    1.0.0.0/24 is subnetted, 1 subnets
O       1.1.1.0 [110/2] via 12.1.1.1, 00:01:30, FastEthernet0/0
    23.0.0.0/24 is subnetted, 1 subnets
C       23.1.1.0 is directly connected, Serial2/0
    8.0.0.0/24 is subnetted, 1 subnets
C       8.8.8.0 is directly connected, FastEthernet1/0
    12.0.0.0/24 is subnetted, 1 subnets
C       12.1.1.0 is directly connected, FastEthernet0/0
R2#
```

```
PC1> ip 1.1.1.1 255.255.255.0 1.1.1.0
Invalid gateway address

PC1> ip 1.1.1.1 255.255.255.0 1.1.1.10
Checking for duplicate address...
PC1 : 1.1.1.1 255.255.255.0 gateway 1.1.1.10

PC1>
PC1>
```

```
PC2> ip 8.8.8.8 255.255.255.0 8.8.8.10
Checking for duplicate address...
PC1 : 8.8.8.8 255.255.255.0 gateway 8.8.8.10

PC2>
```

```
PC1> ping 8.8.8.8
8.8.8.8 icmp_seq=1 timeout
84 bytes from 8.8.8.8 icmp_seq=2 ttl=62 time=72.629 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=62 time=44.246 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=62 time=76.415 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=62 time=55.456 ms

R2#conf t
R2(config)#access-list 100 permit icmp host 1.1.1.1 host 8.8.8.8
R2(config)#access-list 100 permit ip any any
```

```

R2(config)#access-list 101 permit icmp host 1.1.1.1 host 8.8.8.8
R2(config)#access-list 101 permit ip any any
R2(config)#int s2/0
R2(config-if)#ip access-group 100 in
R2(config-if)#exit
R2(config)#int fa0/0
R2(config-if)#ip access-group 101 in
R2(config-if)#exit R2(config)#exit

```

```

R2#sh ip access-list
Extended IP access list 100
  10 permit icmp host 1.1.1.1 host 8.8.8.8
  20 permit ip any any (13 matches)
Extended IP access list 101
  10 permit icmp host 1.1.1.1 host 8.8.8.8
  20 permit ip any any (10 matches)
R2#

```

```

R1#conf t
R1(config)#ip access-list extended s1
R1(config-ext-nacl)#permit icmp host 1.1.1.1 host 8.8.8.8
R1(config-ext-nacl)#exit
R1(config)#route-map
R1 permit 10
R1(config-route-map)#match ip add s1
R1(config-route-map)#set ip next-hop 23.1.1.2
R1(config-route-map)#exit

```

```

R1
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#do sh history
  ip add 23.1.1.1 255.255.255.0
  no shut
  exit
  int f1/0
  ip add 1.1.1.10 255.255.255.0
  no shut
  exit
  router ospf 100
  network 12.1.1.0 0.0.0.255 area 1
  network 23.1.1.0 0.0.0.255 area 1
  network 1.1.1.0 0.0.0.255 area 1
  exit
  ip access-list extended s1
  permit icmp host 1.1.1.1 host 8.8.8.8
  exit
  route-map R1 permit 10
  match ip add s1
  set ip next-hop 23.1.1.2
  exit
  do sh history
R1(config)#

```

```

R1(config)#int f1/0

```


R1(config-if)#ip policy route-map R1

R1(config-if)#do sh history

```
R1#sh route-map
route-map R1, permit, sequence 10
  Match clauses:
    ip address (access-lists): s1
  Set clauses:
    ip next-hop 23.1.1.2
  Policy routing matches: 0 packets, 0 bytes
R1#sh ip policy
Interface      Route map
Fa1/0          R1
R1#
```

```
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  1.0.0.0/24 is subnetted, 1 subnets
C       1.1.1.0 is directly connected, FastEthernet1/0
  23.0.0.0/24 is subnetted, 1 subnets
C       23.1.1.0 is directly connected, Serial2/0
  8.0.0.0/24 is subnetted, 1 subnets
O       8.8.8.0 [110/2] via 12.1.1.2, 00:29:26, FastEthernet0/0
  12.0.0.0/24 is subnetted, 1 subnets
C       12.1.1.0 is directly connected, FastEthernet0/0
R1#
```

```
R2#sh ip access-list
Extended IP access list 100
  10 permit icmp host 1.1.1.1 host 8.8.8.8
  20 permit ip any any (13 matches)
Extended IP access list 101
  10 permit icmp host 1.1.1.1 host 8.8.8.8
  20 permit ip any any (10 matches)
R2#sh ip access-list
Extended IP access list 100
  10 permit icmp host 1.1.1.1 host 8.8.8.8
  20 permit ip any any (132 matches)
Extended IP access list 101
  10 permit icmp host 1.1.1.1 host 8.8.8.8
  20 permit ip any any (129 matches)
R2#
```

Practical 6

Practical 6: Configuring remote SPAN (On Cisco Packet Tracer).



```
Switch#conf t
```

```
Switch(config)#vlan 8
```

```
Switch(config-vlan)#remote-span
```

```
Switch(config-vlan)#monitor session 1 source int fa0/2
```

```
Switch(config)#monitor session 1 destination remote vlan 8 reflector-port fa0/5
```

```
Switch#conf t
```

```
Switch(config)#vlan 8
```

```
Switch(config-vlan)#remote-span
```

```
Switch(config-vlan)#monitor session 1 source remote vlan 8
```

```
Switch(config)#monitor session 1 destination int fa0/2
```

```
Switch#sh monitor session 1
Session 1
-----
Type                : Remote Destination Session
Description          : -
Source Ports        :
  Both              : Fa0/2
Dest RSPAN VLAN     : 8

Switch#show monitor session 1 detail
Session 1
-----
Type                : Remote Destination Session
Description          : -
Source Ports        :
  RX Only           : None
  TX Only           : None
  Both              : Fa0/2
Source VLANs        :
  RX Only           : None
  TX Only           : None
  Both              : None
Source RSPAN VLAN   : None
Destination Ports   : None
Encapsulation       : Native
  Ingress           : Disabled
Filter VLANs        : None
Dest RSPAN VLAN     : 8

Switch#
```

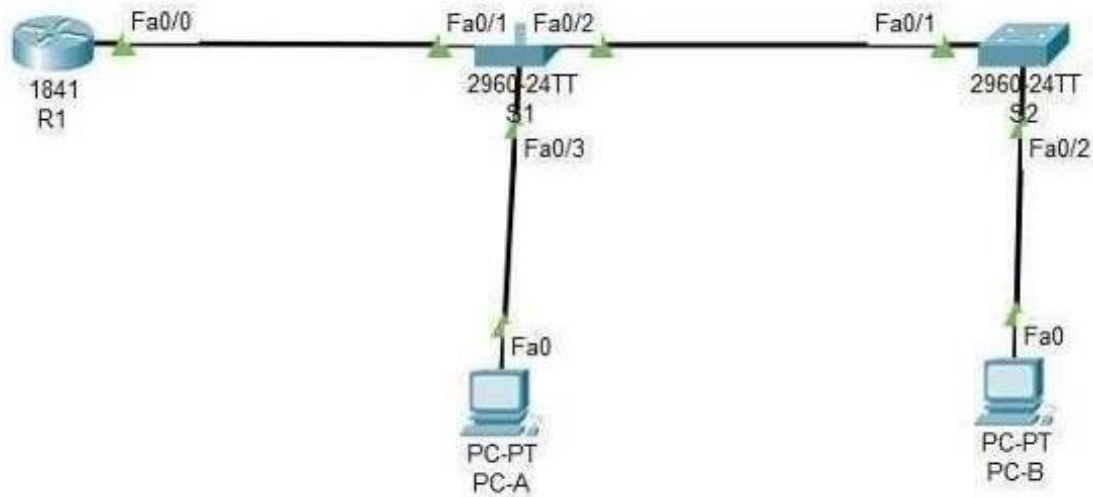
Ctrl+F6 to exit CLI focus.

Copy Paste

☐ Top

Practical 7

Practical 7: Configuring Inter-VLAN Routing(On Packet Tracer).



Device	Interfaces	IP Address	Subnet Mask	Default Gateway
R1	F0/0.10	192.168.10.1	255.255.255.0	N/A
	F0/0.20	192.168.20.1	255.255.255.0	
	F0/0.30	192.168.30.1	255.255.255.0	
	F0/0.1000	N/A	N/A	
S1	VLAN 10	192.168.10.11	255.255.255.0	192.168.10.1
S2	VLAN 10	192.168.10.12	255.255.255.0	192.168.10.1
PC-A	NIC	192.168.20.3	255.255.255.0	192.168.20.1
PC-B	NIC	192.168.30.3	255.255.255.0	192.168.30.1

VLAN	Name	Interface Assigned
10	Management	S1: VLAN 10
		S2: VLAN 10
20	Sales	S1: F0/3
30	Operations	S2: F0/2
999	Parking-Lot	S1: F0/4-24, G0/1-2
		S2: F0/3-24, G0/1-2
1000	Native	N/A

```

Switch#conf t
Switch(config)#hostname S1
S1(config)#vlan 10
S1(config-vlan)#name Management
S1(config-vlan)#exit
S1(config)#vlan 20
S1(config-vlan)#name Sales

```



```

S1(config-vlan)#exit
S1(config)#vlan 30
S1(config-vlan)#name Opearitions
S1(config-vlan)#exit
S1(config)#vlan 999
S1(config-vlan)#name Parking-Lot
S1(config-vlan)#exit
S1(config)#vlan 1000
S1(config-vlan)#name Native
S1(config-vlan)#exit
S1(config)#end

```

```
S1#sh vlan br
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	Management	active	
20	Sales	active	
30	Opearitions	active	
999	Parking-Lot	active	
1000	Native	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
S1#
```

```

S1#conf t
S1(config)#int vlan 10 ip add 192.168.10.11 255.255.255.0
S1(config-if)#exit
S1(config)#ip default-gateway 192.168.10.1
S1(config)#int vlan 10
S1(config-if)#no shut
S1(config-if)#exit
S1(config)#end
S1#conf t
S1(config)#int range f0/4-24, g0/1-2
S1(config-if-range)#switchport mode access

```

S1(config-if-range)#switchport access vlan 999

S1(config-if-range)#shutdown

```
S1#sh vlan br
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3
10	Management	active	
20	Sales	active	
30	Opearations	active	
999	Parking-Lot	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
1000	Native	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
S1#
```

Switch(config)#hostname S2

S2(config)#vlan 10

S2(config-vlan)#name Management

S2(config-vlan)#exit

S2(config)#vlan 20

S2(config-vlan)#name sales

S2(config-vlan)#exit

S2(config)#vlan 30

S2(config-vlan)#name Operations

S2(config-vlan)#exit

S2(config)#vlan 999

S2(config-vlan)#name Parking-Lot

S2(config-vlan)#exit

S2(config)#vlan 1000

S2(config-vlan)#name Native

S2(config-vlan)#exit

S2(config)#end

```

S2#sh vlan br

VLAN Name                Status    Ports
-----
1      default              active    Fa0/1, Fa0/2, Fa0/3,
Fa0/4                      Fa0/5, Fa0/6, Fa0/7,
Fa0/8                      Fa0/9, Fa0/10,
Fa0/11, Fa0/12             Fa0/13, Fa0/14,
Fa0/15, Fa0/16             Fa0/17, Fa0/18,
Fa0/19, Fa0/20             Fa0/21, Fa0/22,
Fa0/23, Fa0/24             Gig0/1, Gig0/2

10     Management           active
20     sales                active
30     Operations           active
999    Parking-Lot          active
1000   Native               active
1002   fddi-default         active
1003   token-ring-default   active
1004   fddinet-default      active
1005   trnet-default         active
S2#

```

S2#conf t

S2(config-if)#ip add 192.168.10.12 255.255.255.0

S2(config-if)#int vlan 10

S2(config-if)#no shut

S2(config-if)#exit

S2(config)#int range f0/3-24,g0/1-2

S2(config-if-range)#switchport mode access

S2(config-if-range)#switchport access vlan 999

S2(config-if-range)#shutdown

```

S2#sh vlan br

VLAN Name                Status    Ports
-----
1      default              active    Fa0/1, Fa0/2
10     Management           active
20     sales                active
30     Operations           active
999    Parking-Lot          active    Fa0/3, Fa0/4, Fa0/5,
Fa0/6                      Fa0/7, Fa0/8, Fa0/9,
Fa0/10                     Fa0/11, Fa0/12,
Fa0/13, Fa0/14             Fa0/15, Fa0/16,
Fa0/17, Fa0/18             Fa0/19, Fa0/20,
Fa0/21, Fa0/22             Fa0/23, Fa0/24,
Gig0/1, Gig0/2

1000   Native               active
1002   fddi-default         active
1003   token-ring-default   active
1004   fddinet-default      active
1005   trnet-default         active
S2#

```

S1#conf t

S1(config-if)#switchport mode access

S1(config-if)#switchport access vlan

20 S1(config-if)#exit

S1(config)#end

```
S1#sh vlan br
```

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2
10 Management	active	
20 Sales	active	Fa0/3
30 Opeartions	active	
999 Parking-Lot	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
1000 Native	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
S1#
```

S2#conf t

S2(config)#int f0/2

S2(config-if)#switchport mode access

S2(config-if)#switchport access vlan 30

S2(config-if)#exit

S2(config)#end

```
S2#sh vlan br
```

VLAN Name	Status	Ports
1 default	active	Fa0/1
10 Management	active	
20 sales	active	
30 Operations	active	Fa0/2
999 Parking-Lot	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
1000 Native	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
S2#
```

```
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk allowed vlan 10,20,30,1000
S1(config-if)#exit
S1(config)#end
```

```
Router#conf t
Router(config)#int f0/0.10
Router(config-subif)#description vlan 10
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip add 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int f0/0.20
Router(config-subif)#description vlan 20
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip add 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int f0/0.30
Router(config-subif)#description vlan 30
Router(config-subif)#ip add 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int f0/0.1000
Router(config-subif)#description NATIVE
Router(config-subif)#encapsulation dot1q 1000 native
Router(config-subif)#exit
Router(config)#int f0/0
Router(config-if)#no shutdown
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10,
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20,
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.30,
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/0.1000, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.1000,
changed state to up
```



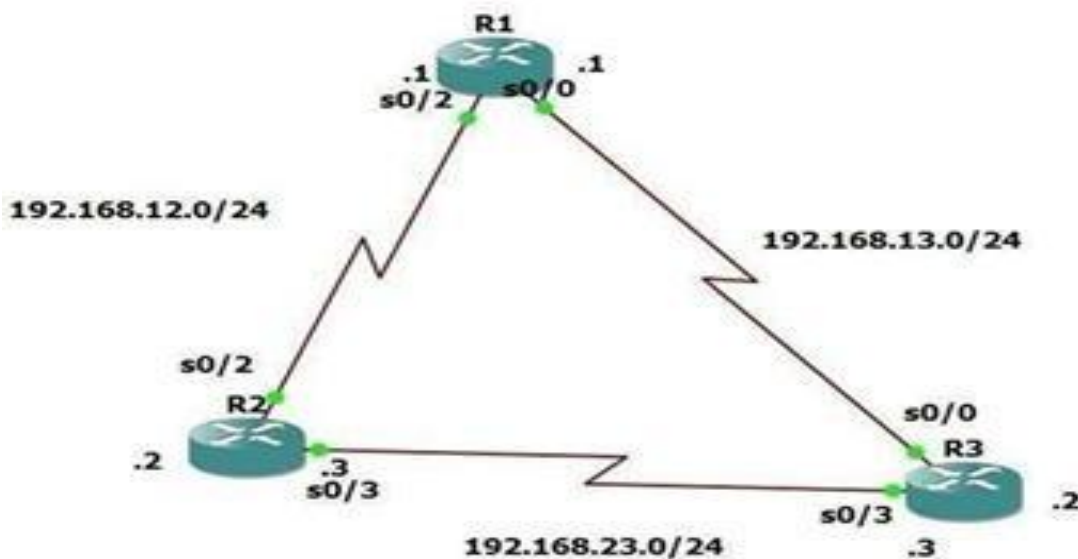
```

Router#sh ip int br
Interface      IP-Address      OK? Method Status
Protocol
FastEthernet0/0 unassigned      YES unset  up
up
FastEthernet0/0.10 192.168.10.1    YES manual up
up
FastEthernet0/0.20 192.168.20.1    YES manual up
up
FastEthernet0/0.30 unassigned      YES unset  up
up
FastEthernet0/0.1000 unassigned      YES unset  up
up
FastEthernet0/1 unassigned      YES unset  administratively
down down
Vlan1 unassigned      YES unset  administratively
down down
Router#

```

Practical 8 A

Practical 8 (A): Configuring MPLS(On GNS3).



```
R1#conf t
R1(config)#int se1/0
R1(config-if)#ip add 192.168.13.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int se1/2
R1(config-if)#ip add 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
```

```
R2#conf t
R2(config)#int se1/2
R2(config-if)#ip add 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int se1/3
R2(config-if)#ip add 192.168.23.1 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
```

```
R3#conf t
R3(config)#int se1/0
R3(config-if)#ip add 192.168.13.2 255.255.255.0
R3(config-if)#no shut
```

```
R3(config-if)#exit
R3(config)#int se1/3
R3(config-if)#ip add 192.168.23.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
```

```
R1#conf t
R1(config)#router ospf 1
R1(config-router)#network 192.168.12.0 255.255.255.0 area 0
R1(config-router)#network 192.168.13.0 255.255.255.0 area 0
R1(config-router)#end
```

```
R2#conf t
R2(config)#router ospf 1
R2(config-router)#network 192.168.12.0 255.255.255.0 area 0
R2(config-router)#network 192.168.23.0 255.255.255.0 area 0
R2(config-router)#end
```

```
R3#conf t
R3(config)#router ospf 1
R3(config-router)#network 192.168.13.0 255.255.255.0 area 0
R3(config-router)#network 192.168.13.0 255.255.255.0 area 0
R3(config-router)#network 192.168.23.0 255.255.255.0 area 0
R3(config-router)#end
```

```
R1#conf t
R1(config)#int se1/0
R1(config-if)#mpls ip
R1(config-if)#int se1/2
R1(config-if)#mpls ip
R2#conf t
R2(config)#int se1/2
R2(config-if)#mpls ip
R2(config-if)#int se1/3
R2(config-if)#mpls ip
R3#conf t
R3(config)#int se1/0
R3(config-if)#mpls ip
R3(config-if)#int se1/3
R3(config-if)#mpls ip
```

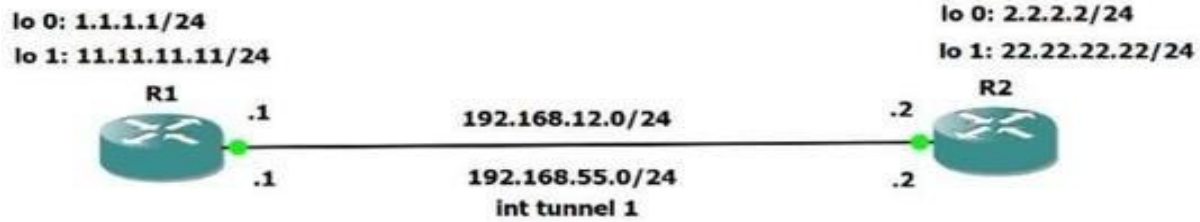
```
R1#sh mpls int
Interface      IP      Tunnel  Operational
Serial1/0      Yes (ldp) No      Yes
Serial1/2      Yes (ldp) No      Yes
R1#
```

```
R2#sh mpls int
Interface      IP      Tunnel  Operational
Serial1/2      Yes (ldp) No      Yes
Serial1/3      Yes (ldp) No      Yes
R2#
```

```
R3#sh mpls int
Interface      IP      Tunnel  Operational
Serial1/0      Yes (ldp) No      Yes
Serial1/3      Yes (ldp) No      Yes
R3#
```


Practical 8 B

Practical 8 (B): Implement VRF.



```
R1#conf t
R1(config)#hostname R1
R1(config)#int fa0/0
R1(config-if)#ip add 192.168.12.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#ip vrf R1
R1(config-vrf)#exit
R1(config)#int lo0
R1(config-if)#int lo0
R1(config-if)#ip vrf forwarding R1
R1(config-if)#ip add 1.1.1.1 255.255.255.0
R1(config-if)#int lo0
R1(config-if)#ip vrf forwarding R1
R1(config-if)#ip add 11.11.11.11 255.255.255.0
```

```
R1
% Invalid input detected at '^' marker.
R1#R1(config-if)#ip add 1.1.1.1 255.255.255.0
% Invalid input detected at '^' marker.
R1#R1(config-if)#int lo0
% Invalid input detected at '^' marker.
R1#R1(config-if)#ip vrf forwarding R1
% Invalid input detected at '^' marker.
R1#R1(config-if)#ip add 11.11.11.11 255.255.255.0
% Invalid input detected at '^' marker.
R1#
% Unknown command or computer name, or unable to find computer address
R1#% Invalid input detected at '^' marker.
% Invalid input detected at '^' marker.
R1#
R1#R1(config-if)#ip add 11.11.11.11 255.255.255.0
% Invalid input detected at '^' marker.
R1#R1(config-if)#2
% Unknown command or computer name, or unable to find computer address
R1#EXIT(R1)#
R1#R1#ip
R1#
R1#sh ip int br
Interface      IP-Address      OK? Method Status    Prot
-----
FastEthernet0/0 192.168.12.1    YES manual up        up
Serial1/0       unassigned      YES unset  administratively down down
Serial1/1       unassigned      YES unset  administratively down down
Serial1/2       unassigned      YES unset  administratively down down
Serial1/3       unassigned      YES unset  administratively down down
LoopBack0      11.11.11.11     YES manual up        up
Tunnel1        192.168.55.1    YES manual up        up
R1#
```

```
R2#conf t
```

```

R2(config)#hostname R2
R2(config)#int fa0/0
R2(config-if)#ip add 192.168.12.2 255.255.255.0
R2(config-if)#no shut

R2(config-if)#
*Jun 24 11:41:01.651: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
R2(config-if)#
*Jun 24 11:41:01.651: %ENTITY_ALARM-6-INFO: CLEAR INFO Fa0/0 Physical Port
Administrative State Down
*Jun 24 11:41:02.651: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R2(config-if)#exit
R2(config)#ip vrf R2
R2(config-vrf)#exit
R2(config)#int lo0
R2(config-if)#ip
*Jun 24 11:41:26.487: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
changed state to up
R2(config-if)#ip vrf forwarding R2
R2(config-if)#ip add 2.2.2.2 255.255.255.0
R2(config-if)#int lo0
R2(config-if)#int lo0
R2(config-if)#ip vrf forwarding R2
R2(config-if)#ip add 22.22.22.22 255.255.255.0

```

```

R2
Connected to Dynamips VM "R2" (ID 2, type c7200) - Console port
Press ENTER to get the prompt.
*Jun 24 11:41:01.651: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
R2(config-if)#ip add 192.168.12.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#
*Jun 24 11:41:01.651: %ENTITY_ALARM-6-INFO: CLEAR INFO Fa0/0 Physical Port
Administrative State Down
*Jun 24 11:41:02.651: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up
R2(config-if)#exit
R2(config)#ip vrf R2
R2(config-vrf)#exit
R2(config)#int lo0
R2(config-if)#ip
*Jun 24 11:41:26.487: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
changed state to up
R2(config-if)#ip vrf forwarding R2
R2(config-if)#ip add 2.2.2.2 255.255.255.0
R2(config-if)#int lo0
R2(config-if)#int lo0
R2(config-if)#ip vrf forwarding R2
R2(config-if)#ip add 22.22.22.22 255.255.255.0
R2#
*Jun 24 11:45:16.411: %SYS-5-CONFIG_I: Configured from console by console
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#router eigrp 12
R2(config-router)#no auto-summary
R2(config-router)#address-family ipv4 vrf R2
R2(config-router-af)#network 2.2.2.0
R2(config-router-af)#network 22.22.22.0
R2(config-router-af)#network 192.168.55.0
R2(config-router-af)#autonomous-system 12
R2(config-router-af)#
*Jun 24 11:50:24.263: %DUAL-5-NBRCHANGE: IP-EIGRP(1) 12: Neighbor 192.168.55.1 (
Tunnel1) is up: new adjacency
R2(config-router-af)#Z
R2#
*Jun 24 11:50:26.371: %SYS-5-CONFIG_I: Configured from console by console
R2#ping vrf R2 11.11.11.11
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 11.11.11.11, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 72/73/76 ms
R2#ping vrf R2 11.11.11.11 source loopback0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 11.11.11.11, timeout is 2 seconds:
Packet sent with a source address of 22.22.22.22
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 72/76/88 ms
R2#
R2#?
R2#?TERM
R2#?conf t
?
? Invalid input detected at '^' marker.
R2#Enter configuration commands, one per line. End with CNTL/Z.
? Invalid input detected at '^' marker.

```

```
R1#conf t
R1(config)#int tunnel 1
R1(config-if)#t
*Jun 24 11:44:54.823: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1,
changed state to down
R1(config-if)#tunnel source fa0/0
R1(config-if)#tunnel destination 192.168.12.2
R1(config-if)#ip v
*Jun 24 11:45:20.927: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1,
changed state to up
R1(config-if)#ip vrf forwarding R1
R1(config-if)#ip add 192.168.55.1 255.255.255.0
```

```
R2#conf
R2(config)#int tunnel 1
R2(config-if)#tunnel source fa0/0
R2(config-if)#tunnel destination 192.168.12.1
R2(config-if)#ip vrf forwarding R2
R2(config-if)#ip add 192.168.55.2 255.255.255.0
```

```
R1#conf t
R1(config)#router eigrp 12
R1(config-router)#no auto-summary
R1(config-router)#address-family ipv4 vrf R1
R1(config-router-af)#network 1.1.1.0
R1(config-router-af)#network 11.11.11.0
R1(config-router-af)#network 192.168.55.0
R1(config-router-af)#autonomous-system 12
```

```
R2#conf t
R2(config)#router eigrp 12
R2(config-router)#no auto-summary
R2(config-router)#address-family ipv4 vrf R2
R2(config-router-af)#network 2.2.2.0
R2(config-router-af)#network 22.22.22.0
R2(config-router-af)#network 192.168.55.0
R2(config-router-af)#autonomous-system 12
```

```
R2#R2 (config-if)#tunnel source Fa0/0
^
% Invalid input detected at '' marker.
R2#R2 (config-if)#tunnel destination 192.168.12.1
^
% Invalid input detected at '' marker.
R2#R2 (config-if)#ip vrf
^
% Invalid input detected at '' marker.
R2#R2 (config-if)#ip vrf forwarding R2
^
% Invalid input detected at '' marker.
R2#R2 (config-if)#ip add 192.168.55.2 255.255.255.0
^
% Invalid input detected at '' marker.
R2#R2 (config-if)#?
% Unknown command or computer name, or unable to find computer address
R2#
R2#ping vrf virar 11.11.11.11

% IP routing table virar does not exist
R2#X(X's)
R2#sh ip int br
Interface
FastEthernet0/0      192.168.12.2      YES manual up      up
Serial1/0             unassigned        YES unset  administratively down down
Serial1/1             unassigned        YES unset  administratively down down
Serial1/2             unassigned        YES unset  administratively down down
Serial1/3             unassigned        YES unset  administratively down down
Loopback0             22.22.22.22      YES manual up      up
Tunnell              192.168.55.2      YES manual up      up
R2#
```